Chien-Yi Yang

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Profile

Computer architecture Ph.D. candidate with research in system architecture definition, performance modeling, power and energy modeling, and hardware/software co-design for AI accelerators.

Education

University of California San Diego **Sept. 2022 – Mar. 2027 (Expected)**
Ph.D. in Computer Science and Engineering

National Taiwan University **Sept. 2017 – June 2021**
B.S. in Electrical Engineering

Research Experience

University of California San Diego **Sept. 2022 – Present**
Graduate Research Assistant

PNMAX | Architecture and Performance Modeling **MICRO 2026 (to appear); First author**

- Developed Python analytical performance models of tensor processing, dataflows, scheduling, memory hierarchy and interconnects, enabling PPA analysis and performance projections for HBM-PIM, UPMEM and AttAcc.
- Validated microarchitecture models against UniNDP, achieving **32.1× faster HBM-PIM model evaluation** with squared Pearson latency correlation above 0.99.
- Performed workload characterization and parameter sensitivity studies for Transformers (Llama-3, GPT-2 and OPT) and other tensor workloads (CNNs, FCs), identifying buffer, memory bandwidth and reduction bottlenecks to guide accelerator architecture exploration and workload mapping.

OptiPIM | PIM Modeling and Evaluation **ISCA 2025; Distinguished Artifact Award**

- Modeled ILP-based workload mapping for PIM inference optimization, quantifying a **72.8× modeled speedup** over the Samsung HBM-PIM heuristic on the evaluated workloads.
- Conducted model validation against cycle-accurate Ramulator-based simulation over **12,000 mappings**, obtaining squared Pearson latency correlations above 0.99 for HBM-PIM and SIMDRAM.

NN-Steiner | Neural Combinatorial Optimization **AAAI 2024**

- Developed PyTorch neural network algorithms for Steiner-tree optimization, reducing mean wirelength by **2.99%** versus FLUTE (A=18) across 100 uniformly sampled 5,000-point instances.
- Implemented distributed training with PyTorch/NCCL and CPU/GPU profiling, enabling multi-GPU model updates and execution traces for training bottleneck analysis.

CSBO | Constrained Hardware/Software Optimization **IEEE ESL 2026**

- Developed constraint-scheduled Bayesian optimization for hardware-aware model optimization and software-hardware co-design over a roughly 500,000-configuration design space in approximately **1.5 hours**.
- Automated energy, latency and silicon area evaluation with Timeloop/CiMLoop and PyTorch accuracy constraints, finding PPA trade-offs with up to **287× lower energy** or 14× lower delay than prior HDnn-PIM.

HD-PIM | Noise-Aware Accelerator Co-Design **ICCAD 2024**

- Optimized HD-PIM through hardware-aware algorithm design and noise-aware Bayesian optimization, identifying a MNIST design with **35.38% lower peak power** than HyperMetric at 3.4× delay and 4% higher area.
- Modeled ASIC pipeline microarchitecture and ReRAM non-idealities, enabling power, performance and area (PPA) comparisons between digital and analog in-memory computation.

Duostra | C++ Quantum Mapping **ACM TQC 2024**

- Developed C++ graph algorithms for qubit routing and scheduling in Duostra, integrated into qsyn, a quantum compilation framework with nearly 200 GitHub stars.
- Benchmarked graph-based mapping on circuits over 50 qubits, reducing mean mapping cost by **21.75%** versus the per-circuit best of four competing mappers.

Industry Experience

MediaTek Research **Dec. 2021 – June 2022**
R&D Engineer Intern *Taipei, Taiwan*

Distributed T5 Training

- Scaled distributed T5 training with PyTorch Distributed and data/model parallelism to **100+ GPUs**.
- Optimized T5 training through workload profiling and critical-path analysis in PyTorch Profiler, achieving **30× higher throughput**.

Technical Skills

Software development: C++; Python; algorithm prototyping; PyTorch; NCCL; PyTorch Profiler.

Architecture: Performance modeling; power and energy modeling; power efficiency; microarchitecture; memory hierarchy; software-hardware co-design; workload characterization; performance analysis; bottleneck analysis.

Deep learning: CNNs and Transformers; neural network operators.

Simulation and optimization: Ramulator; Timeloop; CiMLoop; CACTI; BoTorch.

AI tools: Agentic workflow.

Selected Publications

[1] **Chien-Yi Yang***, Jiantao Liu*, Yue Pan, Tajana Rosing, Minxuan Zhou, Lana Josipović, "PNMAX: Mapping-Architecture Co-Exploration Framework for Processing Near Memory", MICRO 2026 (to appear).

[2] **Chien-Yi Yang**, Flavio Ponzina, Minxuan Zhou, Raid Ayoub, Pietro Mercati, Tajana Rosing, "Constraint-Scheduled Bayesian Optimization for Software-Hardware Co-Optimization on HDnn-PIM", IEEE Embedded Systems Letters (ESL), 2026.

[3] Esha Singh, Dongxia Wu, **Chien-Yi Yang**, Tajana Rosing, Rose Yu, Yian Ma, "Divide and Learn: Multi-Objective Combinatorial Optimization at Scale", ICML 2026.

[4] Jiantao Liu, Minxuan Zhou, Yue Pan, **Chien-Yi Yang**, Lana Josipović, Tajana Rosing, "OptiPIM: Optimizing Processing-in-Memory Acceleration Using Integer Linear Programming", ISCA 2025.

[5] **Chien-Yi Yang**, Minxuan Zhou, Flavio Ponzina, Suraj Sathya Prakash, Raid Ayoub, Pietro Mercati, Mahesh Subedar, Tajana Rosing, "Multi-Objective Software-Hardware Co-Optimization for HD-PIM via Noise-Aware Bayesian Optimization", ICCAD 2024.

[6] Andrew B. Kahng, Robert R. Nerem, Yusu Wang, **Chien-Yi Yang**, "NN-Steiner: A Mixed Neural-Algorithmic Approach for the Rectilinear Steiner Minimum Tree Problem", AAAI 2024.

[7] Chin-Yi Cheng, **Chien-Yi Yang**, Yi-Hsiang Kuo, Ren-Chu Wang, Hao-Chung Cheng, Chung-Yang (Ric) Huang, "Robust Qubit Mapping Algorithm via Double-Source Optimal Routing on Large Quantum Circuits", ACM Transactions on Quantum Computing (TQC), 2024.